

SAHIL MANE

Machine Learning · Computer Vision · Perception · Robotics Software

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PROFILE

Mechatronics engineer and Robotics Systems Engineering MSc candidate (RWTH Aachen) pairing two years of production data-analytics experience with graduate-level machine learning, computer vision, and perception. Comfortable across the applied-ML stack — Python, PyTorch, OpenCV, SQL — from data pipelines and real-time dashboards to CNN-based perception and robotic manipulation. Seeking ML, perception, and robotics-software roles where rigorous data engineering meets embedded robotics.

CORE COMPETENCIES

Machine Learning & CV: PyTorch, CNNs, OpenCV, image classification, feature engineering, model evaluation

Programming: Python, C, SQL, MATLAB/Simulink, LabVIEW

Data & Analytics: Pandas, NumPy, SciPy, Tableau, data pipelines, real-time dashboards

Robotics & Simulation: ROS 2, Gazebo, CoppeliaSim, kinematics, control, path planning

Embedded & Hardware: Arduino, Pixhawk, sensor fusion, GRBL/G-code, SCADA/PLC, Proteus

CAD & Design: SOLIDWORKS, Fusion 360, AutoCAD

Languages: English (C1), German (A2), Hindi (native), Marathi (native)

PROFESSIONAL EXPERIENCE

Decision Analytics Associate

ZS Associates | Pune/Mumbai, India | Sep 2022 – Aug 2024

- Engineered data-acquisition and selection algorithms across primary and secondary sources, building reproducible pipelines in Python (Pandas, NumPy) and SQL serving analytics for [X]+ client engagements.
- Built and maintained live workflows and real-time Tableau dashboards, giving senior stakeholders [X]%-faster access to decision-ready insights.
- Translated ambiguous business problems into data-driven recommendations and corrective strategies, partnering cross-functionally with [X] stakeholder teams.
- Streamlined recurring analysis and reporting on proprietary platforms, cutting manual effort by an estimated [X]%.
- Authored process/methodology documentation and built onboarding pathways for new associates; recognised as Star Performer at the 2023 Team Summit.

Automation Intern

Mumbai Metro Rail Corporation (MMRCL) | Mumbai, India | May 2021 – Jul 2021

- Supported operation of the TBM and transformer-station systems for the Mumbai Metro Line-3 underground tunnelling programme.
- Built ladder-logic simulations of tunnelling and EOT-crane processes in MATLAB/Simulink and RSLogix 500.
- Optimised the HMI of the tunnelling SCADA system for operator clarity and documented TBM automation processes.

SELECTED PROJECTS

Vision-Guided Robotic Fruit Sorter — Python, CoppeliaSim, PyTorch

- Built an automated sorting cell: a CNN classified conveyor-fed fruit (raw / ripe / overripe / rotten) and drove a simulated robot arm to place each item into one of four bins.
- Trained and evaluated the image-classification model on labelled fruit imagery; integrated perception with pick-and-place control.

Robotic Biomimicry of Avian Aerial Locomotion (Owl Drone) — SOLIDWORKS, MATLAB, Pixhawk

- Prototyped a surveillance drone for wildlife-sanctuary protection, fusing a thermal sensor grid with onboard computer vision to detect unsanctioned human activity and alert rangers.

Automated Biometric Entrance System — Python, LabVIEW, sensors

- Developed a pulse/SpO2 screening + identity gate (Arduino + LabVIEW) with Python data logging and historical-baseline comparison, reaching ~87% screening accuracy.

EDUCATION

M.Sc. Robotic Systems Engineering — RWTH Aachen University, Aachen, Germany | Oct 2024 – Present (current grade 1.7)

Relevant coursework: Machine Learning, Computer Vision, Kinematics & Dynamics, Control Theory, Probabilistic / Bayesian methods.

B.Tech. Mechatronics Engineering (Minors: Robotics, IoT) — NMIMS MPSTME, Mumbai, India | 2018 – 2022 (3.84/4.0)

CERTIFICATIONS & ADDITIONAL

- IELTS Academic — English proficiency C1.
- Treasurer, IETE-SF MPSTME (2020–2021) — managed budgets and events for the student chapter.